# Phase Diagrams




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Reference:
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# Phase, Phase Diagrams and Phase Rule

- In the classical definition, a ***phase*** is a physically distinct, chemically homogeneous and mechanically separable region of a system

- The various states of aggregation of matter, namely, the gaseous, the liquid and the solid states, form separate phases

- The gaseous state is always a single phase, as the atoms (or molecules) in the gas are mixed at the atomic (or molecular) level

- A liquid solution is also a single phase; for example, if salt is dissolved in water, the water molecules, the sodium ions and the chlorine ions are mixed at the atomic level in the solution

- A liquid mixture (e.g. oil and water), on the other hand, forms two separate phases as there is no mixing at the molecular level

- In the solid state, different chemical compositions and different crystal structures are possible so that a solid may consist of several phases; for the same composition, different crystal structures represent different phases; a solid solution has the atoms mixed at the atomic level within the unit cell and is therefore a single phase

- ***Phase diagrams*** are maps that give the relationships between phases in equilibrium in a system as a function of temperature, pressure and composition

# Phase, Phase Diagrams and Phase Rule

- The ***components*** of a system may be elements, ions or compounds; they refer to the independent chemical species that comprise the system
- In the ice-water-steam system, the component is $H_2O$
- In the Cu-Ni system, the elements Cu and Ni are the components, whereas in the $Al_2O_3$-$Cr_2O_3$ system, the two oxides can be taken to be components
- In the Fe-C system, iron and graphite can be the components, but it may be often convenient to choose iron and iron carbide ($Fe_3C$) as the components
- The ***variables*** of a system include the two external parameters, temperature and pressure; within the system, there are variables that specify the compositions of the phases present
- Compositions are expressed as weight % (or atom %) so that the number of variables required to specify completely the composition of a phase is (C – 1), where C is the number of components in the system
- If there are P phases in a system, the total number of composition variables is P(C – 1)
- Including the two external variables (pressure and temperature), the total number of variables is P(C – 1) + 2

# Phase, Phase Diagrams and Phase Rule

- Gibbs Phase Rule states that $F = C - P + 2$

  where F is the ***degrees of freedom*** given by the number of independent variables

- The number of degrees of freedom cannot be more than the total number of variables

$$F = C - P + 2 \leq P(C - 1) + 2$$

- When only one phase is present in a system, the degrees of freedom are equal to the total variables; as the number of phases increases, the degrees of freedom decrease

- The degree of freedom cannot be less than zero so that we have an upper limit to the number of phases that can exist in equilibrium in a given system

- For C = 2

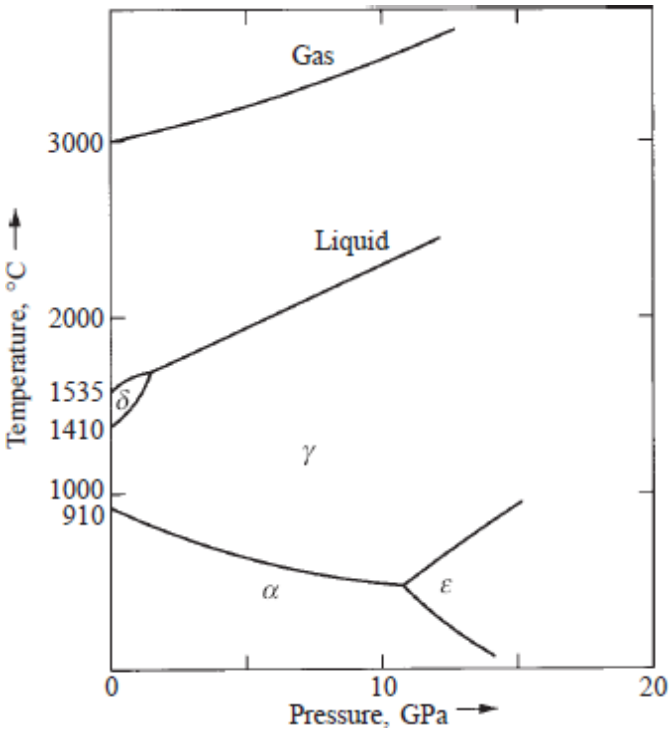
| No. of phases | Total variables $P(C-1)+2$ | Degrees of freedom $C-P+2$ |
|---|---|---|
| 1 | 3 | 3 |
| 2 | 4 | 2 |
| 3 | 5 | 1 |
| 4 | 6 | 0 |

The system cannot have more than 4 phases in equilibrium

# Types of Phase Diagrams

- Phase diagrams are classified on the basis of the number of components in the system
- Single-component systems have unary diagrams, two-component systems have binary diagrams, three-component systems give rise to ternary diagrams, and so on
- In single-component systems, there is no composition variable and the only other variables are temperature and pressure

**Pressure-temperature diagram for the one-component system of iron**

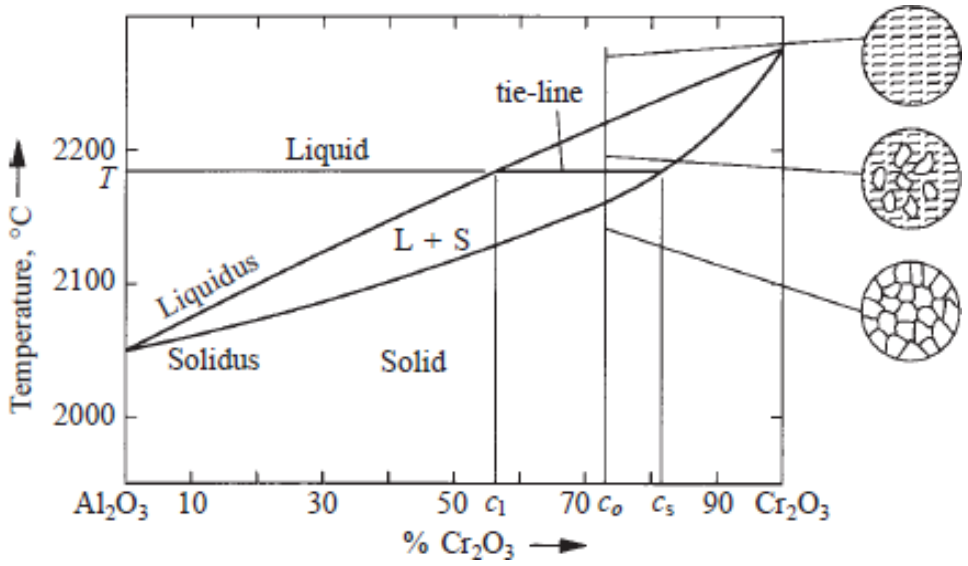


# Types of Phase Diagrams

- Two-component systems have binary phase diagrams; apart from temperature and pressure, there is one composition variable for each of the phases in equilibrium
- We then need a three-dimensional diagram to plot the variations in pressure, temperature and composition;
- In order to simplify the representation of the phase relationships on paper, binary phase diagrams are usually drawn at atmospheric pressure, showing variations in temperature and composition only; the modified phase rule becomes $F = C - P + 1$
- The simplest binary phase diagram is obtained for a system exhibiting complete liquid solubility as well as solid solubility
- The two components dissolve in each other in all proportions both in the liquid and the solid states
- Clearly, the two components must have the same crystal structure besides satisfying the other Hume Rothery's conditions for extensive solid solubility
- Cu-Ni, Ag-Au, Ge-Si and $Al_2O_3$-$Cr_2O_3$ are examples of such systems

# Types of Phase Diagrams

- There are only two phases on the phase diagram, the liquid and the solid phases
- The single-phase regions are separated by a two-phase region (L + S), where both liquid and solid co-exist; in all binary phase diagrams, a two-phase region separates single-phase regions




**The $Al_2O_3$–$Cr_2O_3$ phase diagram; microstructural changes in an overall composition $c_0$ are sketched on the right**

- The phase boundary between the liquid and the two-phase region is called the ***liquidus***, while the boundary between the solid and the two-phase region is called the ***solidus***
- When only one phase is present, the composition axis gives the composition of that phase directly
- When two phases are present, the compositions of the phases are not the same; at the temperature of interest T, a horizontal line called the tie-line is drawn

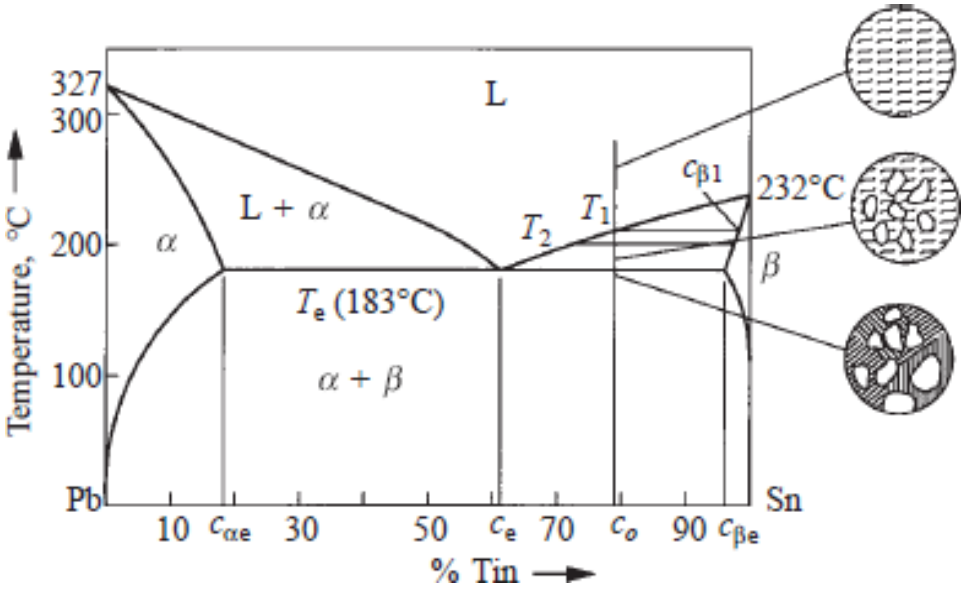
- The points of intersection of the tie-line with the liquidus and the solidus give, respectively, the liquid and the solid compositions, $c_l$ and $c_s$, which are in equilibrium with each other

# Types of Phase Diagrams

- When solid solubility is limited and the melting points of the components are not vastly different, a ***eutectic*** phase diagram usually results

- At the eutectic temperature, the eutectic reaction takes place

$$\text{L} \xleftrightarrow[\text{heating}]{\text{cooling}} \alpha + \beta$$




**The lead–tin phase diagram; microstructural changes for composition $c_o$ are sketched**

- The phase boundary between α and α+β is called the ***solvus***

- The eutectic horizontal is used as a tie-line, the ends of which give the compositions of the α and β phases at this temperature

- On slightly increasing the temperature above $T_e$, either one or both of α and β phases would disappear

- On slight decrease of temperature below $T_e$, the liquid phase would transform to a mixture of α and β

- To denote the zero degree of freedom, the eutectic reaction is called an ***invariant reaction***; the eutectic temperature is known as an ***invariant temperature***

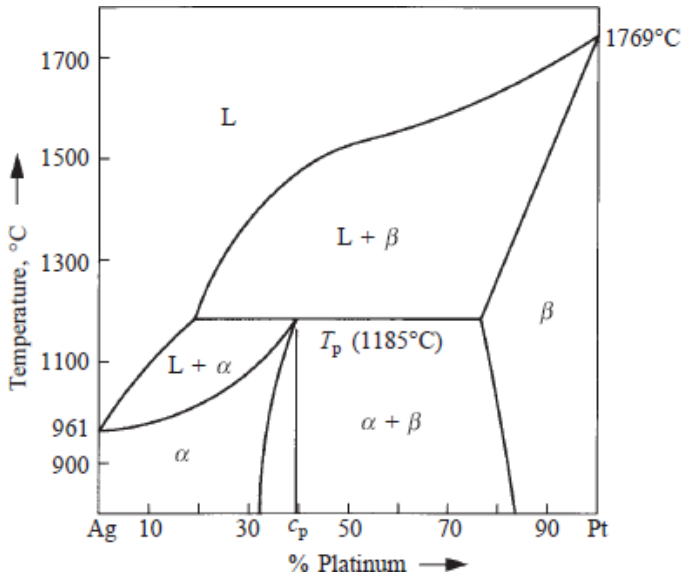
# Types of Phase Diagrams

- A similar invariant reaction occurring entirely in the solid state, where the liquid phase is replaced by a third solid phase γ is called a ***eutectoid*** reaction

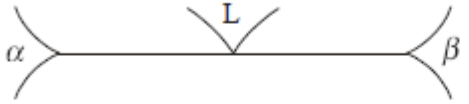
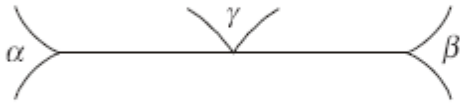
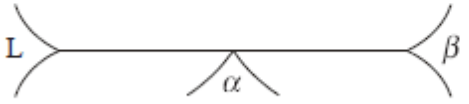
$$\text{cooling} \rightarrow$$
$$\gamma \leftrightarrow \alpha + \beta$$
$$\leftarrow \text{heating}$$

- When the melting points of the components are vastly different from each other, a ***peritectic*** phase diagram may be formed

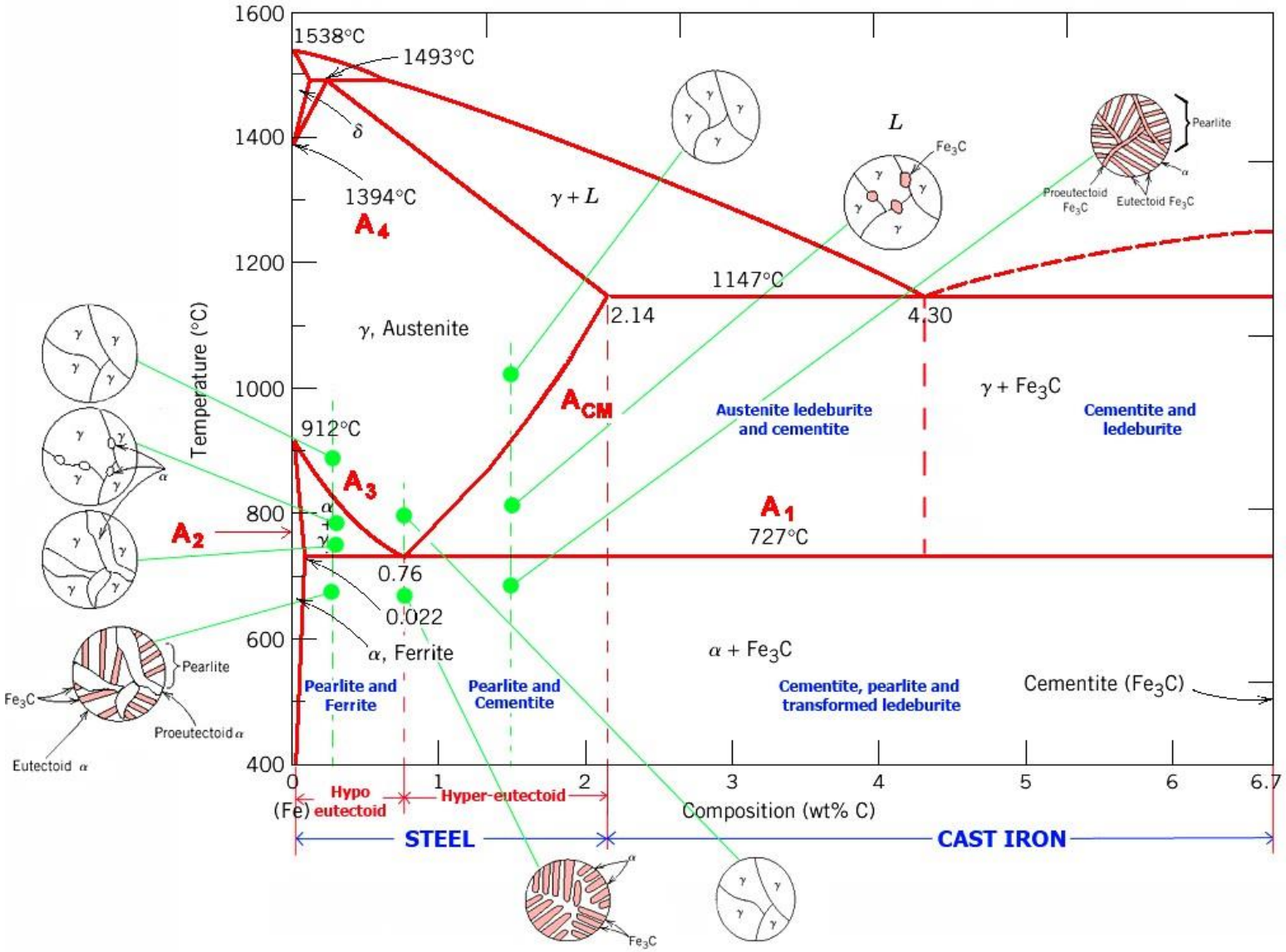
$$\text{cooling} \rightarrow$$
$$L + \beta \leftrightarrow \alpha$$
$$\leftarrow \text{heating}$$



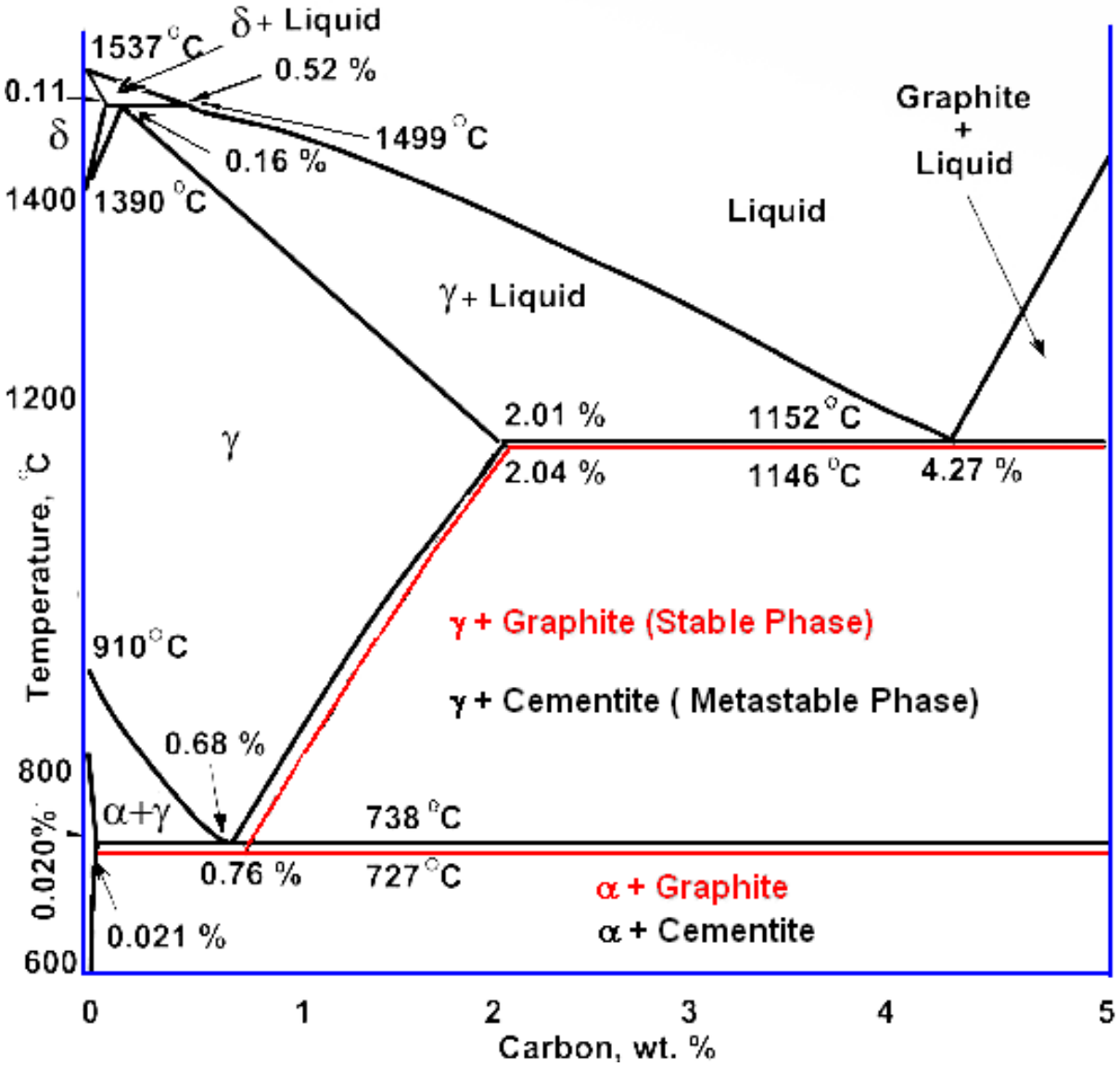

The silver–platinum phase diagram

| Name of the reaction | Details of the reaction | Phase boundaries at the invariant line |
|---|---|---|
| Eutectic | cooling → $L \leftrightarrow \alpha + \beta$ ← heating |  |
| Eutectoid | cooling → $\gamma \leftrightarrow \alpha + \beta$ ← heating |  |
| Peritectic | cooling → $L + \beta \leftrightarrow \alpha$ ← heating |  |
| Peritectoid | cooling → $\gamma + \beta \leftrightarrow \alpha$ ← heating |  |

# Iron-Carbon Phase Diagram



# Iron-Carbon Phase Diagram



# Lever Rule

- ***Lever rule*** derived from mass balance gives the relative amounts of the co-existing phases

- The tie-line is treated as a lever arm, with the fulcrum at the overall composition; for the arm to be horizontal, the weight to be hung at each end must be proportional to the arm length on the other side of the fulcrum. The “weight” at each end corresponds to the amount of the phase at that end

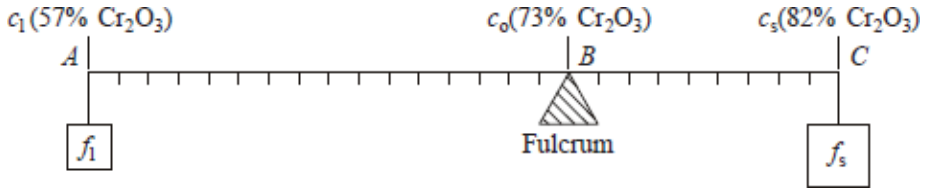



Illustration of the lever rule

weight fractions of liquid and solid as $f_l$ and $f_s$, we obtain

$$f_l = \frac{CB}{CA} = \frac{c_s - c_o}{c_s - c_l} = \frac{82 - 73}{82 - 57} = 0.36$$

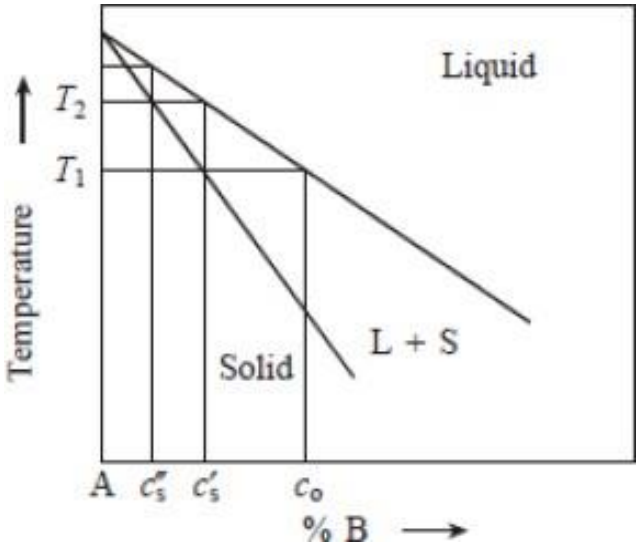
$$f_s = \frac{BA}{CA} = \frac{c_o - c_l}{c_s - c_l} = \frac{73 - 57}{82 - 57} = 0.64$$

- The lever rule cannot be applied at the eutectic or the peritectic temperature, where there are three phases in equilibrium and an isothermal reaction, changing the relative amounts of the phases, can occur

- It can be applied just above or just below the invariant temperature; by using the lever rule, it is possible to estimate (i) the fraction of a proeutectic phase, (ii) the fraction of the eutectic mixture, and (iii) the fraction of a phase that forms part of the eutectic mixture

# Application of phase diagram: Zone refining

- Considering the phase diagram shown below, we need to purify or refine A, which has B as the impurity component
- The ***zone refining*** process is based on the fact that the solid to crystallize first from the melt is usually purer in A than the liquid




- Let the starting composition be $c_0$; this composition starts to solidify at temperature $T_1$
- Just below this temperature, a small quantity of solid of composition $c_s'$, purer in A than $c_o$, separates from the liquid
- Imagine that we stop cooling further, throw away the liquid part and remelt the left-over solid; this small quantity of liquid is now purer, as its composition is $c_s'$
- This overall composition will now start to solidify at a higher temperature $T_2$; again, the first solid to separate will have a composition $c_s''$ purer in A than $c_s'$
- By repeating this sequence of operations a few times, we can get very pure A, even though the quantity of the purified material will be extremely small compared to the starting material

# Application of phase diagram: Zone refining

- In zone refining, this principle of phase separation is used
- The material to be purified is in the form of a long rod
- At any time, only a small length of this rod is melted with the aid of an induction coil or an electron beam
- The coil or beam is moved slowly from one end to the other end of the rod, continuously solidifying the molten zone and remelting fresh material ahead
- Surface tension forces are usually strong enough to hold the molten zone in place without the need for a container, which may contaminate the melt
- If the zone is passed across many times, each time in the same direction, the material at the starting end becomes much purer than the rest of the rod
- In a typical case, ten passes of the molten zone can reduce the impurity level to as low as $10^{-6}$ times the initial value
- This technique to produce ultra high purity materials has enabled us to achieve the sophistication required in the control of impurity concentration in semiconductor crystals